

DISCUSSION OF “DISASTERS, CAPITAL, AND PRODUCTIVITY”

Tarikua Erda

Discussed by Seula Kim

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THIS PAPER

Climate changes + Fixed nature of physical capital

Q: How do floods impact plant capital, productivity, and allocative efficiency in the U.S. manufacturing?

Climate changes + Fixed nature of physical capital

Q: How do floods impact plant capital, productivity, and allocative efficiency in the U.S. manufacturing?

- Use administrative micro data in the U.S. manufacturing sector combined with flood records
 - Detailed data on capital adjustment margins (scrapped, sold, investment, new or used, etc.)
- Adopt an event study design from declared flood events
- Document novel findings on flood impacts on plant entry/exit, capital investment, productivity, and aggregate outcomes
- Draw out mechanism through financing access and policy implications

RECAP: PRODUCTIVITY EFFECTS

Productivity outcomes in post-flood periods:

- Exit ↑ in general, but more for low-TFP plants

Table 2: Plants' 2-year exit likelihood increases after flooding, particularly for lowest performing units

	All plants	Heterogeneity by pre-flood performance	
		TFP	Value added/hour
1.Post flood	0.036*** (0.005)	0.049*** (0.006)	0.048*** (0.006)
1.LowestQuartile		-0.002 (0.007)	-0.007 (0.009)
1.Post flood#1.LowestQuartile		0.011** (0.005)	0.02*** (0.005)
N	422,000	257,000	257,000

RECAP: PRODUCTIVITY EFFECTS

Productivity outcomes in post-flood periods:

- Exit ↑ in general, but more for low-TFP plants
- General effects on labor productivity of plants in flooded areas are not statistically significant

Table 4: Investment and Productivity Outcomes

	Normalized Net Investment	IHS(Net Investment Normalized)	Log(Output/Hour)
Post flood	0.019* (0.0101)	0.015** (0.007)	0.0005 (0.017)
N	66,000	66,000	66,000

RECAP: PRODUCTIVITY AND REALLOCATION EFFECTS

Productivity outcomes in post-flood periods:

- Exit ↑ in general, but more for low-TFP plants
- General effects on labor productivity of plants in flooded areas are not statistically significant
- Labor productivity for plants w/ capital retirement in the flood year ↑ (but not in value added)

Table 7: Flooded plants which replace their capital see higher labor productivity

	Log(Output/hour)	Log(Hourly wage)	IHS(Value added/hour)
Post flood X Retire in flood year	0.045** (0.019)	0.027* (0.014)	0.021 (0.035)
Post flood	-0.022 (0.024)	-0.032* (0.017)	–
Retire in flood year	-0.043 (0.038)	-0.0006 (0.026)	–
N	68,000	68,000	68,000

RECAP: PRODUCTIVITY AND REALLOCATION EFFECTS

Productivity outcomes in post-flood periods:

- Exit $\uparrow$ in general, but more for low-TFP plants
- General effects on labor productivity of plants in flooded areas are not statistically significant
- Labor productivity for plants w/ capital retirement in the flood year $\uparrow$ (but not in value added)
- Aggregate dispersion of output/capital declines; productivity effect is not statistically significant

Table 10: On average, post-flood reallocation enhances productivity

	IQR of $\text{Log}(\text{Output}/\text{Capital stock})$	$\text{Log}(\text{Output}/\text{Hour})$
Post flood	-0.014* (0.009)	0.011 (.016)
N	131,000	21,500

ROBUSTNESS IN PRODUCTIVITY EFFECTS

- Each measure of productivity can pick up different sources:

$$\ln\left(\frac{\text{Output}}{\text{Labor}}\right) = \ln\left(\frac{\text{TVS}}{\text{Deflator}}\right) - \ln(\text{Labor})$$

$$\ln\left(\frac{\text{ValueAdded}}{\text{Labor}}\right) = \ln\left(\frac{\text{TVS} + \Delta\text{Inventory}}{\text{Deflator}}\right) - \ln\left(\frac{\text{EnergyCosts}}{\text{Deflator}_E}\right) - \ln\left(\frac{\text{MaterialCosts}}{\text{Deflator}_M}\right) - \ln(\text{Labor})$$

$$\ln(\text{TFP}) = \ln\left(\frac{\text{TVS} + \Delta\text{Inventory}}{\text{Deflator}}\right) - \alpha_K \ln\left(\frac{\text{CapitalCosts}}{\text{Deflator}_K}\right) - \alpha_E \ln\left(\frac{\text{EnergyCosts}}{\text{Deflator}_E}\right) - \alpha_M \ln\left(\frac{\text{MaterialCosts}}{\text{Deflator}_M}\right) - \alpha_L \ln(\text{Labor})$$

- ⇒ Not consistent use of measures and lacking robustness make it hard to interpret
- Is it true productivity improvement or something else?
- ⇒ Increases labor or VA productivity could be driven by capital deepening, not necessarily indicates improvement in productivity/efficiency (Syverson 2011); Any labor reductions after flooding?
- ⇒ Any effects through price heterogeneity or markups? (Foster et al. 2008, Syverson 2011)

AGGREGATE IMPLICATIONS

- The aggregate results seem subtle and may mask important dynamics
- Would be useful to decompose aggregate productivity changes into entry, exit, within-plant improvements and reallocation to investigate the relative importance and allocative efficiency (Olley-Pakes decomposition or Foster et al. 2001, 2006)

$$\Delta P_{st} = \underbrace{\sum_{f \in C} \omega_{ft-1} \Delta P_{ft}}_{\text{within term}} + \underbrace{\sum_{f \in C} (P_{ft-1} - P_{st-1}) \Delta \omega_{ft}}_{\text{between terms}} + \underbrace{\sum_{f \in C} \Delta P_{ft} \Delta \omega_{ft}}_{\text{cross/covariance terms}} + \underbrace{\sum_{f \in N} \omega_{ft} (P_{ft} - P_{st-1})}_{\text{entry terms}} - \underbrace{\sum_{f \in X} \omega_{ft-1} (P_{ft-1} - P_{st-1})}_{\text{exit terms}}$$

⇒ How is the aggregate change in productivity accounted for by each term?

POST-ENTRY PERFORMANCE AFTER FLOODS

- Positive selection of entrants after floods, but how about their post-entry performance?
- The answer is unclear given the following results:
 - Entrants/young plants invest in second-handed capital after floods
 - The use of second-handed capital does not help for productivity or survival

⇒ How can we evaluate the reallocation of (used) capital from incumbents to young firms after floods? What would be the right policy to help with it?

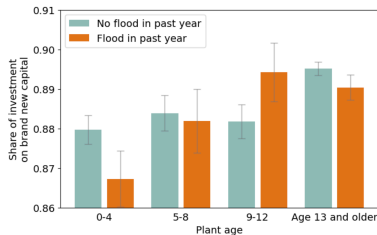


Table 9: Purchase of used capital and young plants' performance

	Entrant Plants (all single & multi-unit firms)			Small Young Firms
	Log(Output/hour) (1)	Survival (Max age observed) (2)		Log(Output/hour) (4)
Post flood	0.026 (0.0534)	0.462 (1.036)	0.883 (1.037)	0.122** (0.051)
Used machine	-0.026 (0.043)	-0.799 (0.618)		0.076 (0.048)
Post flood X Used machine	0.0689 (0.051)	-0.769 (0.799)		-0.101* (0.06)
Used capital			-0.462* (0.608)	
Post flood X Used capital			-1.455* (0.819)	
N	5,300	5,300	5,400	7,800

SOME OTHER COMMENTS

- Contemporaneous capital retirement may not necessarily be a proxy for physical damage
 - It may rather reflect the responsiveness of plants for a given magnitude of shocks
 - Pre-trend pattern exists for their investment
 - The interpretation is likely an upper bound of the damage-replacement channel
- Any heterogeneity across “firms”?
 - The main analysis is conducted primarily at the plant level
 - Examining heterogeneity across “firms” would also be informative
 - All else equal, does the response of plants to the shock vary by firm characteristics?
 - Firm age, size, asset, and the number or geographic distribution of plants may play a role
 - Can potentially use MOPS to see the effect of managerial practices

SOME OTHER COMMENTS (CONT'D)

- Direct evidence for financing mechanism
 - Non-declared events may have differences in flood severity or depend on president discretion
 - Can use the amount of federal assistance and directly interact with plant-level net investment, productivity, as well as the types of capital purchased (new vs used)
 - Any specific impacts on small/young firms?
- What if plants can forecast floods? Any variations across counties depending on flood frequency?
Any relocation decisions of plants?

CONCLUSION

This paper:

- Provides new insights on the intersection of disaster response, capital allocation, and productivity
- Strengths lie in its rich data use, clear identification, and policy relevance

Comments: interesting paper asking an important question w/ great tools!

- Tightening empirical results and having a more thorough discussion of robustness will benefit it
- Explore productivity measures and robustness
- Clarify and strengthen the aggregate implications
- Investigate post-entry performance and the role of firm heterogeneity
- Direct evidence for financing mechanism